# Venkata Midhun Mareedu

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## Professional Summary

Full Stack Developer with **2+ years of experience** building scalable web applications and aspiring AI Engineer focused on production-grade **Generative AI** applications using **Large Language Models (LLMs)**, **RAG (Retrieval-Augmented Generation)**, and prompt engineering. Experienced in integrating AI systems into scalable products with **Next.js**, **FastAPI**, **Vercel AI SDK**, and modern AI toolchains. Passionate about translating machine learning capabilities into reliable, user-facing features.

## Education

**JNTU-GV** — Vijayanagaram, Andhra Pradesh
*Bachelor of Technology, Computer Science & Engineering* — *Jan. 2021 – April 2024*

## Skills

**Languages**: Python, Java, JavaScript, Go, C
**AI/ML**: Generative AI, Large Language Models (LLMs), RAG, Prompt Engineering, Context Engineering, LangChain.js, Vercel AI SDK, Ollama, Vector Databases, MCP (Servers & Clients)
**Frameworks & Libraries**: React.js, Next.js, Electron.js, Node.js, Express.js, FastAPI
**Developer Tools**: Turborepo, Vite, Docker, Kubernetes, Git, GitHub Actions, GitLab CI/CD, Vitest, Playwright
**Cloud & IaC**: AWS (EC2), Google Cloud (Compute Engine), Oracle Cloud (Compute VMs), Terraform, Kubernetes
**UI, State & Forms**: Tailwind CSS, shadcn UI, Framer Motion, TanStack (Query, Router), Zustand, React Hook Form, Zod
**Databases**: MySQL, PostgreSQL, MongoDB, VectorDB (ChromaDB, LanceDB)

## Professional Experience

**SDE-I** — Aug 2025 – Present
*DeeptaAI* — *Hyderabad, India*

- Architected scalable AI Chatbot and Form SDKs using Turborepo for cross-platform university integration with reusable LLM orchestration patterns.
- Engineered academic infrastructure modules to manage complex program and course hierarchies.
- Implemented secure multi-channel payment systems for self-funded and loan-based transactions.
- Developed automated post-exam workflows with high-concurrency result processing and re-correction logic, reducing manual operational overhead.

**Associate SDE** — Dec 2024 – July 2025
*Apxor* — *Hyderabad, India*

- Built a Generative AI chatbot that translated natural language into ClickHouse SQL, enabling automated analytics and data visualization workflows.
- Optimized web application performance to 98% by implementing caching strategies, lighthouse audits, and performance monitoring.
- Led frontend development for high-traffic platforms, optimizing performance and user retention for AI-assisted analytics experiences.

**Full Stack Web Developer** — Feb 2024 – Aug 2024
*IncrivelSoft, Inc* — *Hyderabad, India*

- Co-led the frontend team in delivering responsive MERN stack interfaces and high-fidelity UI components.
- Integrated complex UI with backend APIs to ensure reliable data flows and seamless end-to-end UX.

## Projects

**HireKit** | *Python, FastAPI, PostgreSQL, React, Vite, Docker, Oracle Cloud, LLMs*

- Built an **evidence-backed, anonymized resume screening** platform: every score cites a verified quote, and all LLM calls go through one gateway with token caps, logging, and a USD 8 spend limit.
- Deployed the API, worker, and web app on a single Oracle Cloud VM using Docker, Caddy, and Compose.

**Synapse** | *Electron.js, React, LanceDB, Ollama, Langchain.js, AI SDK*

- Developed a **local AI-enhanced PDF viewer** desktop application with LangChain.js and Vercel AI SDK.
- Implemented built-in AI features including inline question answering, semantic search, and context-aware page redirection.
- Used LanceDB to store embeddings for low-latency retrieval and React PDF Viewer for core document interaction workflows.

**Support AI Chatbot** | *React, MongoDB, AI SDK, Go lang*

- Built a **RAG-driven support chatbot widget + server** (NPM package + script) with configurable system prompts and pluggable tools for enterprise knowledge bases.

**Overmind** | *Skill.md, LaTex*

- Overmind is a LaTeX-based resume workflow for creating a clean base resume and generating role-specific variants quickly.
- Supports ATS-oriented tuning in Cursor/VS Code with reproducible PDF output.